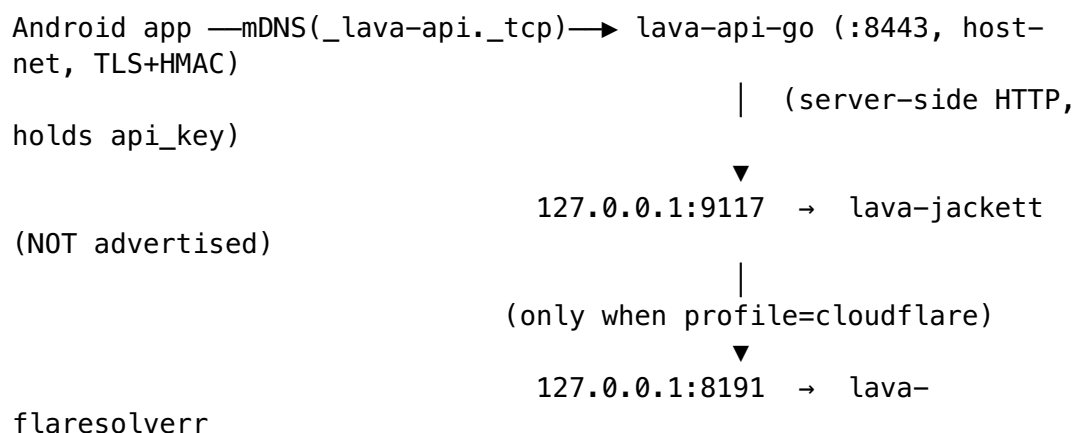


Jackett Torznab Sidecar — Deployment & Validation Guide

Status: deployment-ready (compose fragment + `.env.example` keys + validation script authored; no container booted in this authoring pass).
Companion: [jackett-local-stack-research.md](#) (decision dossier) and [jackett-local-stack-implementation.md](#) (Go Torznab client). Operator decision implemented: **Jackett as a Torznab sidecar in the Lava local stack, fronted by lava-api-go, mDNS-discovered, NOT bundled in the APK.**

This guide covers how to start the sidecar, validate it end-to-end with one command, perform the one-time (stateful, partly manual) Jackett indexer setup for IPTorrents (with FlareSolvrr), and how the Android app reaches it.

1. Topology



- The app discovers **exactly one** service type — `_lava-api._tcp` (= lava-api-go, default `LAVA_API_MDNS_TYPE` in lava-api-go/internal/discovery/mdns.go). Jackett has no mDNS advertisement and is **never** put on the LAN.
 - lava-api-go runs `network_mode: host` (mDNS requires host net). A host-net container **cannot** resolve the bridge DNS name lava-jackett, so the sidecar's port is published to the **host loopback** (127.0.0.1:9117), and lava-api-go reaches it at `http://127.0.0.1:9117`. Loopback bind keeps it off the LAN (same posture as lava-postgres's 127.0.0.1:8432).
 - **FlareSolvrr** (headless Chromium) is only needed for Cloudflare-protected trackers (notably **IPTorrents**). It is heavy, so it lives behind the `cloudflare` profile and is OFF by default. Jackett reaches it over the lava-net bridge at `http://lava-flaresolvrr:8191` (both are bridge members); the loopback publish of 8191 is only for the validation curl.
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2. Files

File	Role
tools/lava-containers/ docker-compose.jackett.yml	compose fragment — lava-jackett (+ optional lava-flaresolVERR) on lava-net, loopback-published
scripts/validate-jackett-sidecar.sh	one-command end-to-end readiness check (bring up → wait healthy → probe caps + flaresolVERR → teardown)
.env.example	placeholders for all LAVA_API_JACKETT_* + LAVA_JACKETT_* + LAVA_FLARESOLVERR_* keys
lava-api-go/internal/ jackett/	the Go Torznab client lava-api-go uses (see implementation doc)
lava-api-go/internal/ config/config.go	reads LAVA_API_JACKETT_* env (the consumer side)

3. Environment variables (§6.R — nothing hardcoded)

Copy .env.example → .env and fill these (the .env is gitignored):

Variable	Consumed by	Purpose / default
LAVA_API_JACKETT_ENABLED	lava-api-go	turn the Jackett route on (false)
LAVA_API_JACKETT_URL	lava-api-go	http:// 127.0.0.1:9117 (loopback — NOT a bridge name; see §1)
LAVA_API_JACKETT_APIKEY	lava-api-go + healthcheck + validation script	the Jackett api_key (no default — generated on first run)
LAVA_API_JACKETT_DEFAULT_INDEXER	lava-api-go + healthcheck	indexer id for the caps probe (all)
LAVA_JACKETT_IMAGE	compose	Jackett image — pin by digest in real .env
LAVA_FLARESOLVERR_IMAGE	compose	FlareSolVERR image — pin by digest
LAVA_JACKETT_BIND_HOST / LAVA_JACKETT_HOST_PORT	compose compose	loopback bind (127.0.0.1 / 9117)

Variable	Consumed by	Purpose / default
LAVA_FLARESOLVERR_BIND_HOST / LAVA_FLARESOLVERR_HOST_PORT		loopback bind (127.0.0.1 / 8191)
LAVA_JACKETT_CONFIG_DIR	compose	gitignored / config host volume (./.jackett-config) — §6.H secrets store
LAVA_JACKETT_PUID / _PGID / _TZ	compose	LinuxServer user/ group/timezone
LAVA_FLARESOLVERR_LOG_LEVEL	compose	FlareSolvrr verbosity (info)

The `api_key` is the SAME value for the Go service AND the container healthcheck AND the validation script — there is no parallel name to keep in sync. It is a §6.H secret: server-side only, never committed (`.gitignore` excludes `.jackett-config/`), never sent to the device.

4. One-command end-to-end validation

After the `api_key` is set in `.env` (see §5), the main agent runs:

```
# Jackett only (RuTracker / RuTor / NNMClub indexers – no
  Cloudflare):
```

```
scripts/validate-jackett-sidecar.sh
```

```
# Jackett + FlareSolvrr (the IPTorrents / Cloudflare path):
```

```
scripts/validate-jackett-sidecar.sh --cloudflare
```

The script auto-detects podman (preferred) then docker, brings the sidecar up, waits for the compose healthchecks to go HEALTHY, then independently re-probes from the host (the §6.B/§6.J discipline — not trusting `State==running`):

- **Jackett:** `GET .../results/torznab/api?t=caps&apikey=...` → expects HTTP 200 + parseable Torznab XML, `api_key` not rejected. (Verbatim body printed.)
- **FlareSolvrr** (`--cloudflare`): `POST /v1 {"cmd":"sessions.list"}` → expects HTTP 200 + `"status":"ok"`. (Verbatim body printed.)

Exit 0 = READY, 1 = a probe failed (logs tailed), 2 = config error (no runtime / missing `api_key`). The stack is torn down on exit unless `--keep`.

5. One-time Jackett setup (stateful — partly MANUAL)

Jackett's configuration lives in the gitignored /config volume and persists across restarts. These steps cannot be fully automated (the dossier's Open Question Q3 — admin HTTP vs per-/config JSON — is unresolved):

5.1 Generate the api_key (required before validation passes)

1. Start Jackett once with the config volume mounted (e.g. the validation script with `--keep`, or `docker compose -f tools/lava-containers/docker-compose.jackett.yml --profile jackett up -d` after the network exists). On first run with an empty /config, Jackett writes a fresh `api_key` into `{LAVA_JACKETT_CONFIG_DIR}/Jackett/ServerConfig.json` under the `APIKey` field (or read it from the WebUI top-right "API Key" field at `http://127.0.0.1:9117`).
2. Copy that value into `.env` as `LAVA_API_JACKETT_APIKEY=...`
3. Re-run the validation script — the caps probe now authenticates. On a zero-indexer config the caps body is an empty `<caps/>`; the script accepts this as "surface live + api_key valid". A real search needs §5.2.

5.2 Add the IPTorrents indexer (with FlareSolverr) — MANUAL WebUI step

IPTorrents sits behind Cloudflare, so FlareSolverr MUST be running and Jackett MUST be pointed at it:

1. Bring up the `cloudflare` profile so `lava-flaresolverr` is running.
2. In the Jackett WebUI (`http://127.0.0.1:9117`) → top-right **gear** / **Settings**: set **FlareSolverr API URL** to `http://lava-flaresolverr:8191` (bridge DNS name — Jackett reaches FlareSolverr over `lava-net`). Save.
3. **Add indexer** → search "IPTorrents" → + → enter the IPTorrents credentials (these are a §6.H secret; they live ONLY in the gitignored /config volume, never in tracked source). Save.
4. (Optional) Add RuTracker / RuTor / NNMClub indexers the same way — they do not need FlareSolverr, so the `jackett-only` profile suffices for them.

Per §6.A: if the indexer-add is ever automated via Jackett's admin HTTP API, that automation MUST gain a contract test asserting the request/flag shape (mirroring `lava-api-go/tests/contract/healthcheck_contract_test.go`) and pin the Jackett version, because the admin API surface is version-sensitive.

6. How the Android app discovers + uses it

The app never sees Jackett. The flow:

1. App auto-discovers `lava-api-go` via mDNS `_lava-api._tcp` (existing discovery; unchanged by this work).

2. App calls lava-api-go's Torznab-backed search route over TLS+HMAC.
3. lava-api-go (holding the api_key server-side) calls Jackett at `http://127.0.0.1:9117/api/v2.0/indexers/<id>/results/torznab/api`, parses the Torznab XML via `internal/jackett`, and returns clean results to the app.
4. Downloads: Jackett answers `/dl/` links with HTTP 302 → `magnet:`; the Go client captures the Location verbatim (does NOT auto-follow) and returns a real magnet or .torrent bytes (see the implementation doc).

Mapping to IPTorrentsConfig

The IPTorrents provider plugin (`core/tracker/iptorrents/.../IPTorrentsConfig.kt`) resolves the **lava-api-go sidecar** base URL (NOT Jackett directly — the app talks only to lava-api-go), in priority order, from:

1. an explicit override,
2. the `iptorrentsJackettBaseUrl` JVM system property (set by the §6.G test task),
3. the `IPTORRENTS_JACKETT_BASE_URL` environment variable (the runtime value).

There is **no compile-time default** (§6.R): unconfigured → `resolve()` returns null → the feature fails honestly ("sidecar base URL not configured") rather than guessing a host. On Android the production value is injected via `BuildConfig` / runtime config in `:core:tracker:client`. So the chain is:

`IPTORRENTS_JACKETT_BASE_URL` → `IPTorrentsConfig.resolve()` → lava-api-go (Torznab route)

→ Jackett (`LAVA_API_JACKETT_URL`)

→ IPTorrents (via FlareSolverr)

7. What CANNOT be automated (honest manual list)

Step	Why it's manual	Mitigation
api_key into .env (§5.1)	Jackett mints the key on first run into its stateful <code>/config</code> ; there is no pre-seeding	validation script refuses the placeholder and prints the exact remediation
IPTorrents indexer add + credentials (§5.2)	per-indexer config is WebUI/API state in <code>/config</code> ; IPTorrents creds are §6.H secrets that must not enter tracked source	documented WebUI steps; future §6.A-contract-tested automation is owed (dossier Q3)
FlareSolverr URL setting in Jackett (§5.2)	a Jackett WebUI/config-file setting, not an env var Jackett reads	one-time WebUI step; persisted in <code>/config</code>
Image digest pinning		

Step	Why it's manual	Mitigation
	the real digest is environment-specific; <code>.env.example</code> carries the <code>:latest</code> placeholder	pin by digest in the real <code>.env</code> (§6.K) before any gate run

Everything else — bring-up, healthcheck wait, caps + FlareSolverr probing, teardown — is fully automated by `scripts/validate-jackett-sidecar.sh`.